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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

IV Semester B. E. Supplementary Examinations Oct/Nov-2024

Industrial Engineering and Management

STATISTICS FOR DATA ANALYTICS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables and formulae handbook permitted.

PART-A

			M	BT	CO
1	1.1	Differentiate between descriptive and inferential statistics.	02	2	1
	1.2	What is the central limit theorem in statistics?	02	1	2
	1.3	State any two assumptions of Binomial Distribution.	02	2	2
	1.4	State any two properties of Normal Distribution.	02	2	1
	1.5	What is the standard error of an estimator in statistics?	02	2	2
	1.6	Define Chi-squared test in statistics.	02	1	2
	1.7	Differentiate between correlation and causation.	02	1	1
	1.8	What is the significance of correlation?	02	2	1
	1.9	Define Type I and Type II error.	02	2	2
	1.10	What is critical value?	02	1	2

PART-B

2	a	A part selected for testing is equally likely to have been produced on any one of six cutting tools. (i) What is the sample space? (ii) What is the probability that the part is from tool 1? (iii) What is the probability that the part is from tool 3 or tool 5? (iv) What is the probability that the part is not from tool 4?	08	2	2
	b	A lot of 100 semiconductor chips contains 20 that are defective. (i) Two are selected, at random, without replacement, from the lot. Determine the probability that the second chip selected is defective. (ii) Three are selected, at random, without replacement, from the lot. Determine the probability that all are defective.	08	2	1
3	a	The phone lines to an airline reservation system are occupied 40% of the time. Assume that the events that the lines are occupied on successive calls are independent. Assume that 10 calls are placed to the airline. (i) What is the probability that for exactly three calls the lines are occupied? (ii) What is the probability that for at least one call the lines are not occupied? (iii) What is the expected number of calls in which the lines are all occupied?	08	2	1

	b	Printed circuit cards are placed in a functional test after being populated with semiconductor chips. A lot contains 140 cards, and 20 are selected without replacement for functional testing. (i) If 20 cards are defective, what is the probability that at least 1 defective card is in the sample? (ii) If 5 cards are defective, what is the probability that at least 1 defective card appears in the sample?	08	2	2
		OR			
4	a	The number of failures of a testing instrument from contamination particles on the product is a Poisson random variable with a mean of 0.02 failure per hour. (i) What is the probability that the instrument does not fail in an 8-hour shift? (ii) What is the probability of at least one failure in a 24-hour day?	08	3	3
	b	The number of telephone calls that arrive at a phone exchange is often modeled as a Poisson random variable. Assume that on the average there are 10 calls per hour. (i) What is the probability that there are exactly 5 calls in one hour? (ii) What is the probability that there are 3 or less calls in one hour? (iii) What is the probability that there are exactly 15 calls in two hours? (iv) What is the probability that there are exactly 5 calls in 30 minutes?	08	3	2
5	a	The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with a mean of 12.4 fluid ounces and a standard deviation of 0.1 fluid ounce. (i) What is the probability a fill volume is less than 12 fluid ounces? (ii) If all cans less than 12.1 or greater than 12.6 ounces are scrapped, what proportion of cans is scrapped? (iii) Determine specifications that are symmetric about the mean that include 99% of all cans.	12	4	4
	b	List any four characteristics of normal distribution curve.	04	2	3
		OR			
6	a	The time between the arrival of electronic messages at your computer is exponentially distributed with a mean of two hours. i) What is the probability that you do not receive a message during a two-hour period? ii) If you have not had a message in the last four hours, what is the probability that you do not receive a message in the next two hours? iii) What is the expected time between your fifth and sixth messages?	08	3	4
	b	The time between calls to a corporate office is exponentially distributed with a mean of 10 minutes. i) What is the probability that there are more than three calls in one-half hour? ii) What is the probability that there are no calls within one half hour? iii) Determine x such that the probability that there are no calls within x hours is 0.01.	08	4	4

7	The following data for the dependent variable, y, and the independent variable, x, have been collected using simple random sampling:					

x	10	14	16	12	20	18	16	14	16	18
y	120	130	170	150	200	180	190	150	160	200

i) Construct a scatter plot for these data. Based on the scatter plot, how would you describe the relationship between the two variables?

ii) Compute the correlation coefficient. — 16 | 3 | 3

OR

8 An article in the Journal of Sound and Vibration (Vol. 151.1991, pp.383 – 394) describe a study investigating the relationship between noise exposure and hypertension. The following data are representative of those reported in the article.

y	1	0	1	2	5	1	4	6	2	3
x	60	63	65	70	70	70	80	90	80	80

y	5	4	6	8	4	5	7	9	7	6
x	85	89	90	90	90	90	94	100	100	100

i) Draw a scatter diagram of y (blood pressure rise in millimeter of mercury) versus x (sound pressure level in decibels). Does a simple linear regression model seem reasonable in this situation?

ii) Fit the simple linear regression model using least squares. Find an estimate of σ^2.

iii) Find the predicted mean rise in blood pressure level associated with a sound pressure level of 85 decibals. — 16 | 3 | 3

9 a Explain the General Procedure for Hypothesis Testing. — 08 | 2 | 2

b The mean water temperature downstream from a power plant cooling tower discharge pipe should be no more than 100°F. Past experience has indicated that the standard deviation of temperature is 2°F. The water temperature is measured on nine randomly chosen days, and the average temperature is found to be 98°F.

i) Should the water temperature be judged acceptable with $\alpha = 0.05$?

ii) What is the P-value for this test?

iii) What is the probability of accepting the null hypothesis at $\alpha = 0.05$ if the water has a true mean temperature of 104°F? — 08 | 3 | 3

OR

10 Grades in a statics course and operations research course taken simultaneously were as follows for a group of students.

Statistics Grade	*Operation Research Grade*			
	A	*B*	*C*	*Other*
A	25	6	17	13
B	17	16	15	6
C	18	4	18	10
Others	10	8	11	20

Are the grades in statistics and operations research related? Use $\alpha = 0.01$ in reaching your conclusion. What is the p-value for this test.? — 16 | 3 | 4

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IV Semester B. E. Examinations Oct-2023

Industrial Engineering and Management

STATISTICS FOR DATA ANALYTICS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

1.1	Differentiate between descriptive and inferential statistics.	02
1.2	What is the central limit theorem in statistics?	02
1.3	State any two assumptions of Binomial Distribution.	02
1.4	State any two properties of Normal Distribution.	02
1.5	What is the standard error of an estimator in statistics?	02
1.6	Define Chi-squared test in statistics.	02
1.7	Differentiate between correlation and causation.	02
1.8	What is the significance of correlation?	02
1.9	Define Type I and Type II error?	02
1.10	What is critical value?	02

PART-B

2	a	A part selected for testing is equally likely to have been produced on any one of six cutting tools. i) What is the sample space? ii) What is the probability that the part is from tool 1? iii) What is the probability that the part is from tool 3 and tool 5? iv) What is the probability that the part is not from tool 4?	08
	b	A lot of 100 semiconductor chips contains 20 that are defective. i) Two are selected, at random, without replacement, from the lot. Determine the probability that the second chip selected in defective. ii) Three are selected, at random, without replacement, from the lot. Determine the probability that all are defective.	08
3	a	The phone lines to an airline reservation system are occupied 40% of the time. Assume that the events that the lines are occupied on successive calls are independent. Assume that 10 calls are placed to the airline. i) What is probability that for exactly three calls the lines are occupied? ii) What is probability that for at least one call the lines are not occupied? iii) What is expected number of calls in which the lines are all occupied?	08

	b	Printed circuit cads are placed in a functional test after being populated with semiconductor chips. A lot contains 140 cards, and 20 are selected without replacement for functional testing. i) If 20 cards defective, what is the probability that at least 1 defective card is in the sample? ii) If 5 cards are defective, what is the probability that at least 1 defective card appears in the sample?	08
		OR	
4	a	The number of failures of a testing instrument from contamination particles on the product is a Poisson random variable with a mean of 0.02 failure per hour. i) What is the probability that the instrument does not fail in an 8 – *hour* shift? ii) What is the probability of at least one failure in a 24 – *hour* day?	08
	b	The number of telephone calls that arrive at a phone exchange is often modeled as a Poisson random variable. Assume that on the average there are 10 calls per hour. i) What is the probability that there are exactly 5 calls in one hour? ii) What is the probability that there are 3 or less class in one hour? iii) What is the probability that there are exactly 15 calls in two hours? iv) What is the probability that there are exactly 5 calls in 30 minutes?	08
5	a	The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with a mean 12.4 fluid ounces and a standard deviation of 0.1 fluid ounce. i) What is the probability a fill volume is less than 12 fluid ounces? ii) If all cans less than 12.1 or greater than 12.6 ounces are scrapped, what proportion of cans is scrapped? iii) Determine specifications that are symmetric about the mean that include 99% of all cans.	12
	b	List any four characteristics of normal distribution curve.	04
		OR	
6	a	The time between the arrival of electronic messages at your computer is exponentially distributed with a mean of two hours. i) What is the probability that you do not receive a message during a two-hour period? ii) If you have not had a message in the last four hours, what is the probability you do not receive a message in the next two hours? iii) What is the expected time between your fifth and sixth messages?	08

	b	The time between calls to a corporate office is exponentially distributed with a mean of 10 minutes. i) What is the probability that there are more than three calls in one-half hour? ii) What is the probability that there are no calls within one half hour? iii) Determine x such that the probability that there no calls within x hours is 0.01.	08
7		The following data for the dependent variable, y, and the independent variable, x, have been collected using simple random sampling:	

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OR

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y	5	4	6	8	4	5	7	9	7	6
x	85	89	90	90	90	90	94	100	100	100

i) Draw a scatter diagram of y (blood pressure rise in millimeters of mercury) versus x (sound pressure level in decibel). Does a simple linear regression model seem reasonable in this situation?
ii) Fit the simple linear regression model using least squares. Find and estimate of σ^2
iii) Find the predicted mean rise in blood pressure level associated with a sound pressure level of 85 decibels. 16

9	a	Explain the general procedure for Hypothesis Tests.	08
	b	The mean water temperature downstream from a power plant cooling tower discharge pipe should be no more than $100°F$. Past experience has indicated that the standard deviation of temperature is $2°F$. The water temperature is measured on nine randomly chosen days, and the average temperature is found to be $98°F$. i) Should the water temperature be judged acceptable with $\alpha = 0.05$? ii) What is the P-value for this test? iii) What is the probability of accepting the null hypothesis at $\alpha = 0.05$ if the water has a true mean temperature of $104°F$?	08

OR

10	Grades in a statistics course and an operations research course taken simultaneously were as follows for a group of students. Are the grades in statistics and operations research related? Use $\alpha = 0.01$ in reaching your conclusion. What is the P-value for this test?	16

Statistics Grade	*Operation Research Grade*			
	A	*B*	*C*	*Other*
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B	17	16	15	6
C	18	4	18	10
Other	10	8	11	20

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RV COLLEGE OF ENGINEERING®

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IV Semester B. E. Examinations October-2023

Industrial Engineering and Management

CAD/CAM AND ROBOTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

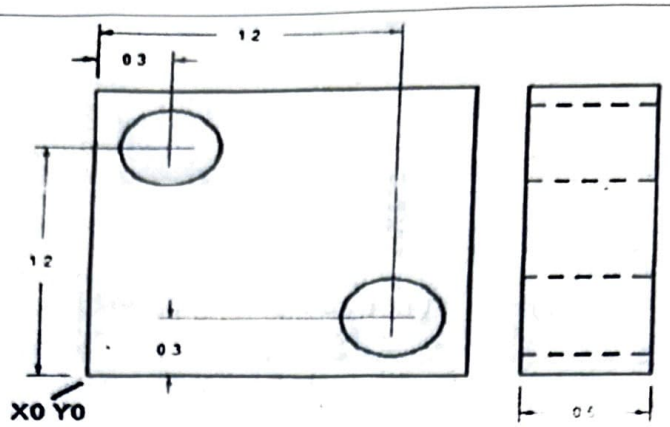
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

1	1.1	Define Robot Programming.	02
	1.2	List any two advantages of suction grippers.	02
	1.3	Identify any two objectives of an Industrial Robot.	02
	1.4	If given an opportunity to build a robot to be used in the department of *IEM* at *RVCE*, what problem would the robot solve and what would be its functionality?	02
	1.5	_____ and _____ are special purpose tool which are used to facilitate production (machining, assembling and inspection operations) when work piece is based on the concept of interchangeability according to which every part will be produced within an established tolerance.	02
	1.6	Students who carry out productivity studies at industries as part of major project sometimes tend to conclude that one or set of robots can be used to automate the entire process and enhance productivity by multifold and cost of commissioning it can be recovered in a very short period by compensating the salaries of the employee who can be removed from the process. Review such inference from the standpoint of India as a country.	02
	1.7	Identify two disadvantages of automation.	02
	1.8	Identify two applications of robots in industrial sector and service sector.	02
	1.9	Tactile sensors are also called as _____ and are sensitive to _____.	02
	1.10	State the Zeroth Law of robotics.	02

PART-B

2	a	Explain the Shigley model with a block diagram.	10
	b	Enumerate any six benefits of *CAD* in automation industry.	06
3	a	Write a *CNC* peck drilling program for the profile given in Fig 3a. The cutting tool used is slot drill cutter and the depth of cut is 5*mm*.	

		 Fig 3a	10
	b	Discuss the various elements of a typical Numerical Control System.	06
		OR	
4	a	Explain with the help of an example i) Linear interpolation ii) Circular interpolation.	06
	b	Explain any five types of Distributed Numerical Control Systems.	10
5	a	Identify major socio-economic considerations favoring automation.	08
	b	With example, state the difference between mechanization and automation.	08
		OR	
6	a	Explain how low-cost automation is achieved in manufacturing industry.	10
	b	Describe six major issues for Automation in Factory Operations.	06
7	a	The robot has to do the following functions. Recommend the types of sensors it should be fitted with to carry out the operations successfully. Also mention one specification of the recommended sensor. Functionality 1: A binary output signal when it comes in contact with an object. Functionality 2: To sense and indicate presence of any object without physical contact. Functionality 3: To identify the distance of sensor from the object. Functionality 4: To estimate the speed with which a manipulator is moved.	08
	b	Explain the advantages and disadvantages of vacuum/suction grippers.	08
		OR	
8	a	Explain the characteristics of an industrial robot.	10
	b	How are the robots classified? Explain.	06
9	a	Assume that the robot must pick up objects from a chute and place them in successive boxes. Choose the sequence of robot activity for which robot programming will have to be written at a later stage.	08
	b	Explain any two robot programming techniques.	08
		OR	
10	a	Explain the factors considered for selection of gripper.	10
	b	Write short notes on types of sensors in robots.	06

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IV Semester B. E. Examinations Oct/Nov-2023

Industrial Engineering and Management

MARKETING MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

1	1.1	List any 2 principles of digital marketing.	02
	1.2	Differentiate between market segmentation and market mix.	02
	1.3	Define Hashtag. List the importance of Hashtag in social media marketing.	02
	1.4	What is keyword research in SEO?	02
	1.5	Define "Dimensions" and "Metrics" in Google analytics.	02
	1.6	List any two key components of a successful email marketing campaign.	02
	1.7	List any two aspects of branding and consistency in web design.	02
	1.8	List any two differences between Mobile Web marketing and Mobile App marketing.	02
	1.9	What is the role of AIDAS framework in the conversion optimization?	02
	1.10	What is the impact of user behavior analysis on website optimization?	02

PART-B

2	a	What is Digital marketing? Explain the differences between Traditional Marketing and Digital Marketing. Highlight any 5 key differences.	08
	b	List and explain various pricing strategies used in marketing management with suitable examples.	08
3	a	Discuss the key metrics used to measure the success of a social media marketing campaign and explain the importance of it for evaluating performance.	08
	b	Enumerate briefly on the best practices for creating compelling and shareable content on social media platforms to increase brand Visibility and Engagement.	08
		OR	
4	a	Explain the concept of Search Engine optimization and its significance in digital marketing.	08
	b	Describe the role of content in SEO. How can business create high quality SEO friendly content?	08

5	a	Discuss the concept of "Acquisition channels" in Google Analytics. How can business use this information to optimize their marketing efforts.	08
	b	Explain the significance of A/B testing and conversion rate optimization in Web Analytics. How can these practices help improve website performance.	08
		OR	
6	a	Describe the difference between transactional emails and promotional emails. Provide examples of each and explain when to use them.	08
	b	List and explain the various reports that can be generated using google analytics.	08
7	a	What is the role of user Interface and user Experience designs in web design for marketing? How can businesses ensure a positive user Journey on their websites?	08
	b	Describe the importance of testing and optimization in web design for Marketing. What tools and methods can be used to analyze and improve Website performance?	08
		OR	
8	a	Discuss briefly on Push Notification in Mobile Marketing. Explain how can businesses use them to engage and retain Mobile App users? Provide examples of effective push notification strategies.	08
	b	Explain the significance of mobile analytics in mobile marketing campaigns. What key metrics should marketers track to measure the success of their mobile strategies?	08
9	a	Describe the conversion funnel in digital marketing. What are the key stages and how can business identify and address drop offs in the funnel?	08
	b	Explain the concept of social proof and its impact on conversion rates. Provide examples of how business can use social proof effectively in marketing.	08
		OR	
10	a	Define Digital Analytics. Describe the key metrics and KPIs used in digital analytics. How can these metrics help business measure the success of their online campaigns?	10
	b	Describe the process of setting up and tracking goals in google analytics. How can businesses use goal tracking to measure website performance.	06